Sense-Plan-Act Pipeline for the IASYS Roadwork Challenge

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1 Introduction

In the IASYS Roadwork Challenge, a vehicle in a 2D ROS2 Jazzy simulator starts at (5, -1.75) m. It must obey a STOP/GO panel before the entry line x = 25m, and pass two static work discs (x ≈ 43-47 m) in the opposing lane. It must then return to its own lane before x = 68 m and stop at the goal (x ≥ 80 m, speed < 0.15 m/s).

This delivery covers **S1** (GO throughout) and **S2** (STOP for t < 10 s, then GO; no crossing of x = 25m during STOP).

2 System architecture

The solution is three independent C++ executables in `roadwork_student`, connected only by ROS topics (Fig. 1).

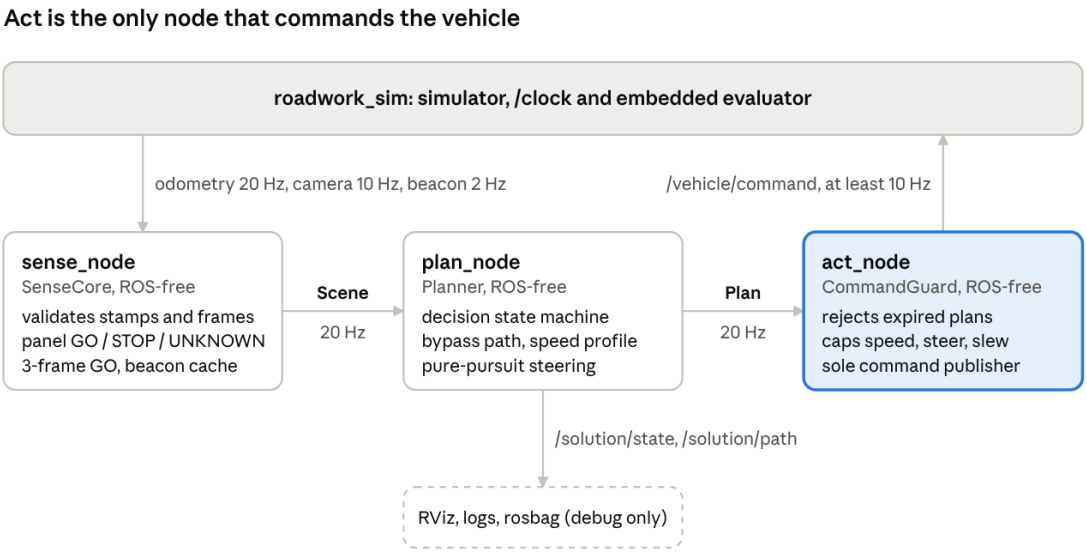


Fig. 1 · system architecture

Each node wraps a ROS-free core (SenseCore, Planner, CommandGuard in include/roadwork_student/), so the same code is unit- and closed-loop tested without ROS. The chain is event-driven at 20 Hz (odometry > scene > plan > command). A 50 ms timer in each node publishes a safe output (invalid scene, hold plan, braking command) when its input stops.

3 Interfaces

Every boundary carries source stamps and an explicit expiry in simulation time. Sense subscribes to odometry, camera and beacon with SensorDataQoS for the best-effort sensors, since a reliable-only subscriber would not match them. LiDAR and radar are unused in D1. The internal topics `/solution/scene` and `/solution/plan` are reliable with depth 1, because only the newest message matters.

The student package `roadwork_student_interfaces` defines two messages:

- **Scene** (sense > plan) carries the ego state with `odom_stamp` and validity, the raw and filtered panel class (UNKNOWN/GO/STOP) with `camera_stamp`, and the cached beacon geometry with `beacon_stamp`. Each block keeps its original sensor stamp.
- **Plan** (plan > act) carries the targets (speed, steering), a speed cap, a hold flag, the state and reason, the stamps used and `valid_until` = stamp + 0.25 s.

Check	Node	Threshold	Action
Odometry age	Sense, Plan	0.20 s	SAFE_STOP
Camera age	Sense	0.30 s	panel > UNKNOWN
GO frame gap	Sense	0.15 s	GO count restarts
Future stamp	all	now + 0.06 s (one clock tick + slack)	sample rejected
Scene age	Plan	0.25 s	hold plan
Plan expiry / age	Act	`valid_until`, 0.30 s	speed 0, steering held

4 Design decisions

The guiding rule is asymmetric: evidence that permits motion (GO) must be confirmed and fresh, while evidence that forbids it (STOP, missing or stale data) acts at once.

4.1 Perception

The panel is a synthetic 96×64 crop with a centered disc and ±18 noise, so a hand-written classifier suffices. Inside a 14 px disc, a pixel counts as red or green if its dominant channel is ≥ 110 and ≥ 70 above both others. A frame ≥ 80 % green is GO, ≥ 80 % red is STOP, otherwise UNKNOWN. Frames with >10 % color in the outer ring are rejected. STOP acts at once, GO needs three consecutive GO frames ≤ 0.15 s apart. Valid beacons are cached, so the path follows work_shift without scenario ids.

4.2 Decisions

Plan's state machine (Fig. 2) holds in WAIT_GO at the start pose, 20 m before entry, until GO is confirmed, which removes any braking-distance risk. GO is then latched and the path frozen, so a noisy frame cannot trigger re-planning.

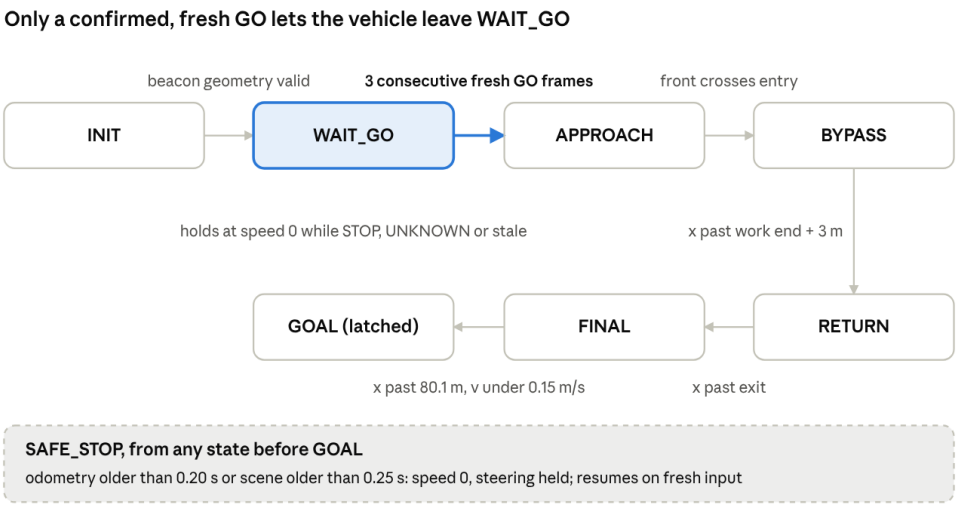


Fig. 2 · Plan decision state machine

4.3 Motion and command guard

The path leaves the lane (y = -1.75 m) at entry -0.5m, holds the bypass (y = +1.25 m) from work start -2m to work end +3m and returns by exit -2.5m. The smootherstep $p(s) = 10s^3 - 15s^4 + 6s^5$ keeps curvature continuous (peak steering 0.215 rad). Speed is the minimum of 3.5 m/s cruise, a curvature limit 15 m ahead, 1.0 m/s² goal braking and the beacon limit -0.3m/s; steering is pure pursuit with a 3-6 m look-ahead. Act brakes on any missing, non-finite, future or expired plan and caps speed (3.8 m/s), steering (±0.45 rad) and steering rate (0.6 rad/s).

5 Experimental setup and results

Regarding S1 episodes, all three seeds finished in 25.00-25.05 s (clearance 1.167 m, peak 3.50 m/s), 0.30-0.35 s slower than closed loop because GO latched only when the first 2 Hz beacon arrived (0.50-0.55 s). The 1-5 rejected commands per run are start-up ticks.

Closed loop (27 runs each)	Pass	Time (s)	Min clearance (m)	Max |CTE| (m)	First motion (s)
S1, no delay	27/27	24.70–24.85	1.156–1.179	0.295	0.20
S1, +0.1 s delay	27/27	24.75–24.90	1.141–1.163	0.255	0.20
S2, no delay	27/27	34.70–34.85	1.156–1.179	0.295	10.20
S2, +0.1 s delay	27/27	34.75–34.90	1.141–1.163	0.255	10.20

In S2 the vehicle stayed at x = 5.0 m throughout STOP and moved 0.20 s after the panel turned GO (the third GO frame). After that, it reproduced the S1 trace exactly (Fig. 3). The extra 0.1 s of command delay cost at most 0.026 m of clearance. The suite therefore covers 12 distinct conditions, not 108.

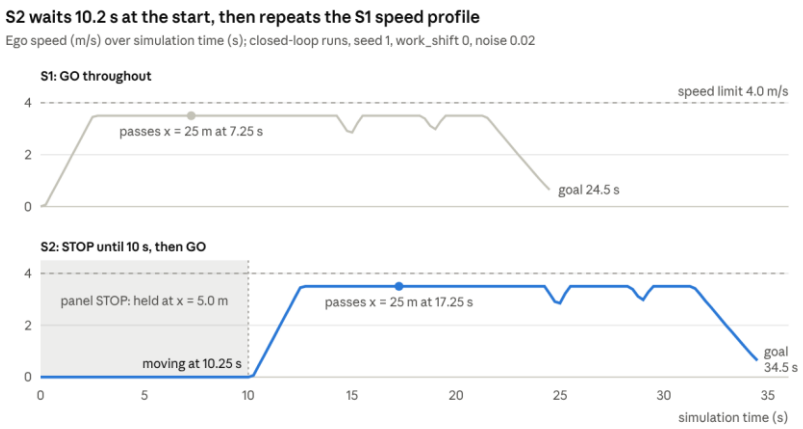


Fig. 3 · S1 and S2 seed 1

6 Discussion and limitations

The pipeline meets S1–S2 in every tested condition, but four gaps remain.

- **Commitment rules.** Entry requires fresh camera evidence but accepts a cached beacon of any age, and the latched GO would ignore a GO > STOP change before the entry line. Neither occurs in S1 and S2, but S4 needs a beacon younger than 1.2 s at entry.
- **Conservative hold.** In S2 the vehicle waits at its start pose rather than at a stop line. This is safe, but adds 20 m of driving after GO.

For D2, Sense will add LiDAR and radar. Plan will add an entry commitment rule, a radar gap check (S3), evidence-loss recovery (S4) and obstacle braking (S5).

7 Conclusion

The pipeline completes S1 and S2 without violations. Next comes the LiDAR clearance, then, for D2, a radar gap check (S3), evidence-loss recovery (S4) and obstacle braking (S5).